



**Ghulam Ishaq Khan Institute of Engineering Sciences
and Technology (GIKI)**

Project Scope Document

for

Project S.A.G.E.: Smart Automated Geriatric Eye

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by

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Abstract

This project presents a comprehensive autonomous Edge-AI monitoring system, S.A.G.E (Smart Automated Geriatric Eye), aimed at addressing critical challenges in geriatric care through intelligent, real-time health analysis. The system is specifically designed to operate in a privacy-preserving manner by processing standard surveillance camera footage locally, eliminating the need for intrusive wearable devices or cloud-based video streaming. While the system is designed for general surveillance integration, a NO-IR camera is utilized specifically during the testing and calibration phase to ensure the model's reliability across all lighting conditions, including total darkness. The solution is deployed on the NVIDIA Jetson MCU platform, which enables high-performance local processing with minimal computational overhead through a resource-efficient state-machine architecture. To achieve high accuracy tailored to the specific demo environment and local needs, the system is trained on a custom and curated dataset. By leveraging TensorRT-optimized YOLOv8 and pose estimation techniques, the prototype delivers actionable health insights—including medication adherence, fall detection, and sleep quality assessment—while strictly maintaining patient privacy.

1. Introduction

S.A.G.E. is a non-intrusive monitoring system designed to enhance geriatric care through intelligent edge-based processing. Utilizing the NVIDIA Jetson MCU as its core computational platform, the system captures and processes footage locally, ensuring strict data privacy. The system interprets complex behaviors such as sleep patterns and fall incidents by integrating optimized deep learning models like YOLOv8. TensorRT optimization is employed to achieve inference speeds of 15+ FPS on the Jetson MCU. While designed to analyze standard surveillance feeds, a NO-IR camera is employed during the testing portion to calibrate the model for diverse lighting environments. Furthermore, the system relies on a custom-curated dataset to ensure the models are specifically tuned for the target geriatric scenarios. A state-machine architecture dynamically activates monitoring modules—such as medication tracking during the day and sleep analysis at night—to reduce computational overhead.

2. Problem Statement

The rapid growth of the elderly population has intensified the need for reliable, continuous, and non-intrusive healthcare monitoring systems. Existing solutions, such as wearable devices and manual caregiving, present significant limitations. Wearables often suffer from low user compliance, as elderly individuals may forget, refuse, or incorrectly use them, leading to inconsistent and unreliable data collection. Manual supervision, while effective, is resource-intensive and not scalable for long-term care.

Alternatively, vision-based monitoring systems that rely on cloud infrastructure introduce serious privacy and security concerns. Continuous video streaming and remote storage of sensitive data compromise patient confidentiality and increase dependency on stable network connectivity. Furthermore, traditional sensor-based systems, such as motion or pressure sensors, lack contextual awareness and fail to accurately interpret complex human activities like medication intake, abnormal movements, or sleep disturbances.

These limitations highlight the need for a solution that ensures accurate, real-time monitoring while maintaining user privacy and minimizing intrusiveness. Additionally, such a system must operate efficiently on low-power, resource-constrained hardware to remain cost-effective and practical.

This project addresses these challenges by developing an edge-based AI monitoring system capable of locally analyzing visual data to detect critical events, ensuring privacy, reliability, and scalability in geriatric care.

3. Problem Solution/Objectives of the Proposed System

The proposed system, S.A.G.E., provides an edge-based AI monitoring solution designed for real-time, non-intrusive, and privacy-preserving care for elderly individuals. By processing all visual data locally on the **NVIDIA Jetson MCU**, the system eliminates dependence on cloud services and ensures data security.

3.1 Objectives

BO-1: Enable real-time monitoring of the elderly by processing visual data locally on Edge-AI Microcontroller.

BO-2: Ensure patient data privacy and security by eliminating external data sharing and performing all analysis directly on-device, and notified through the user interface.

BO-3: Accurately detect critical events such as falls detection.

BO-4: Monitor medication adherence by verifying intake events.

BO-5: Assess sleep quality using infrared imaging and motion-based analysis during nighttime conditions.

BO-6: Optimizing performance and achieving best results on resource-constrained hardware..

BO-7: Develop a cost-effective and scalable prototype using readily available components suitable for practical deployment in real-world geriatric care environments.

BO-8: Utilize a custom and curated dataset specifically developed for this model to improve detection accuracy in geriatric care settings.

BO-9: Validate system reliability across all lighting conditions during the testing phase using a camera.

BO-10: Developing an easy to use, user friendly interface that seamlessly coordinates the hardware and the software.

4. Related System Analysis/Literature Review

Table 1 : Related System Analysis with proposed project solution

Application Name	Features	Weakness	Relevance with the Proposed System – How these weaknesses are addressed
Simple Sensor Motors	Basic movement detection	Lack of context regarding the specific activity	S.A.G.E utilizes YOLOv8 and Pose Estimation to provide rich context, such as identifying if a movement is a fall or medication intake.
Human/Video Surveillance	High accuracy in activity recognition	- Significant lack of privacy - Requires constant human monitoring	S.A.G.E processes all video locally on the Jetson Nano without cloud streaming,

			ensuring total patient privacy.
Cloud-Based AI Monitoring	Advanced behavioral analysis	- Latency issues in reporting critical events - Security risks associated with streaming sensitive data	Security risks associated with streaming sensitive data

5. Vision Statement

The vision of S.A.G.E. (Smart Automated Geriatric Eye) is to transform elderly care through the integration of intelligent, privacy-preserving, and accessible Edge-AI technologies. The system aspires to create a future where continuous health monitoring is seamless, non-intrusive, and respectful of individual dignity, eliminating the need for uncomfortable wearables or invasive surveillance practices. By leveraging on-device intelligence, S.A.G.E. envisions a healthcare ecosystem that prioritizes data security while delivering real-time, context-aware insights into the daily well-being of elderly individuals.

The long-term goal is to develop a scalable and cost-effective solution that can be widely adopted across homes, assisted living facilities, and healthcare institutions, particularly in resource-constrained environments. S.A.G.E. aims to empower caregivers with actionable information, enabling timely interventions and improving overall quality of life for patients. Furthermore, the project envisions expanding its capabilities to incorporate additional health indicators and predictive analytics, ultimately contributing to a smarter, more proactive approach to geriatric care through the advancement of Edge-AI systems.

6. Scope

S.A.G.E. is an autonomous monitoring system designed to process visual data locally via the NVIDIA Jetson MCU. The system is designed to integrate with standard surveillance camera footage; however, for testing and calibration purposes, an IMX219-NOIR camera is utilized to ensure the system handles all lighting conditions effectively.

Key features within the scope include:

- **Custom Dataset Integration:** The model utilizes a self-curated and custom dataset to ensure high-fidelity recognition of geriatric-specific activities.
- **Hardware Optimization:** Deployment on NVIDIA Jetson MCU using TensorRT-optimized YOLOv8 engines to maintain processing speeds over 15 FPS.
- **Medication Adherence:** Detects intake by correlating medicine containers with hand-to-mouth gestures.

- **Fall Detection:** Identifies falls in real-time through skeletal velocity tracking.
- **Sleep Quality:** Calculates a "Restlessness Index" using infrared-based motion history validated during the NOIR camera testing phase.
- **Privacy Logging:** Stores all event data in a local SQLite database.

The system is designed for cost-effective deployment in home or clinical environments. The initial scope focuses on autonomous behavior interpretation through edge AI, utilizing optimized TensorRT engines to maintain a processing speed of over 15 FPS.

7. Project Stakeholders and Roles

Table 2: Project Stakeholders for [PROJECT]

Project Sponsor	[GIKI]
Stakeholder	• Fatima Liaquat, Hussain Munir, Sanawar Khan • Dr. Muhammad Hanif • Senior Design Project Evaluation Panel

8. References

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